EXPERIMENT - 06

DEVELOP A LOW-FIDELITY PROTOTYPE AND CONVERT IT INTO DIGITAL WIREFRAMES USING INKSCAPE

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Aim

To design and develop a detailed low-fidelity prototype for a banking application and digitally convert the prototype into wireframes using Inkscape.
The purpose of the prototype is to visualize the application layout, navigation structure, screen flow, and user interactions before actual coding and implementation. The low-fidelity prototype helps in identifying usability issues, improving user experience, and validating the system design at an early stage of development.

Procedure

1. Requirement Collection and Analysis

- Studied the requirements of a basic banking application.
- Identified the essential features that should be available for users.
- The following modules were selected for the prototype:

Main Modules

1. User Login and Authentication
2. User Registration
3. Dashboard/Home Screen
4. Account Balance View
5. Money Transfer
6. Transaction History
7. Bill Payment
8. Notifications
9. Profile Management
10. Logout Option

- Determined the target users and their basic needs such as:

    - Easy navigation
    - Quick access to banking services
    - Simple and secure user interface

2. Planning the User Interface Structure

- Planned the navigation flow between screens.
- Decided how users would move from one screen to another.
- Created a screen hierarchy for the application.

Example Navigation Flow

- Login → Dashboard
- Dashboard → Balance View
- Dashboard → Transfer Money
- Dashboard → Transaction History
- Dashboard → Profile Settings
- Identified common UI elements required in each screen:
    - Header section
    - Navigation menu
    - Buttons
    - Text fields
    - Icons
    - Cards
    - Lists

3. Creating Hand-Drawn Low-Fidelity Sketches

- Drew rough sketches of all screens on paper.
- Used simple shapes and placeholders instead of actual graphics.
- Focused on functionality and layout rather than colors and styling.

Screens Sketched

- Login Page
- Registration Page
- Home Dashboard
- Balance Checking Page
- Fund Transfer Page
- Transaction History Page
- Profile Page

Design Considerations

- Proper alignment of components
- Easy readability
- Logical placement of buttons
- Simple navigation structure
- Minimal user confusion

4. Evaluating the Paper Prototype

- Reviewed the sketches to check:
    - Navigation clarity
    - Button placement
    - User friendliness
    - Consistency across screens
- Identified possible usability issues such as:
    - Overcrowded screens
    - Confusing navigation paths
    - Missing action buttons
- Modified the sketches wherever necessary before digitization.

5. Opening and Configuring Inkscape

- Installed and launched the software.

- Created a new document workspace.
- Set appropriate page dimensions for mobile application screens.
- Enabled grid and alignment tools for accurate placement.

6. Converting Sketches into Digital Wireframes

Creating Layout Elements

Used the following tools in Inkscape:

- Rectangle Tool:
    - For buttons
    - Input fields
    - Cards
    - Navigation bars
- Text Tool:
    - For labels
    - Screen titles
    - Button names
    - Placeholder text
- Line and Shape Tools:
    - For separators
    - Navigation indicators
    - Layout divisions

Designing Individual Screens

Login Screen

- Added username and password fields
- Added login button
- Included "Forgot Password" option

Dashboard Screen

- Added account summary section

- Included quick action buttons
- Displayed navigation menu

Money Transfer Screen

- Added account number field
- Added transfer amount field
- Included submit button

Transaction History Screen

- Created list placeholders for transaction records
- Added filtering and scrolling indicators

Profile Screen

- Added user details section
- Included edit profile option
- Added logout button

7. Maintaining Low-Fidelity Characteristics

- Used grayscale colors only.
- Avoided detailed icons and images.
- Focused only on structure and functionality.
- Used placeholder text instead of real data.
- Maintained simple and minimal design.

8. Alignment and Consistency Checking

- Checked spacing between UI components.
- Ensured uniform button sizes and font usage.
- Verified alignment of text fields and menus.
- Maintained consistency across all screens.

9. Exporting the Wireframes

- Saved the editable project file.
- Exported the wireframes into:
    - PNG format for image viewing
    - PDF format for documentation and presentation
- Organized all screens properly for future reference and development.

Output

Result

A detailed low-fidelity prototype for the banking application was successfully designed and converted into digital wireframes using Inkscape.

The developed wireframes clearly represented:

- Application structure
- Screen layouts
- Navigation flow
- User interaction process
- Core banking functionalities

The prototype helped in visualizing the complete banking application before implementation and can be further used for:

- Usability testing

- User feedback collection
- UI/UX improvements
- High-fidelity prototype development
- Final software implementation stages